

Task 1: Dataset Description

1.1 Characteristics

The UCI dataset contains 101,766 clinical encounters across 130 US hospitals (1999–2008), described by 50 columns.[1] After accounting for identifiers (`encounter_id`, `patient_nbr`) and the target `readmitted`, 47 input features are available for modeling, yielding a sample-to-feature ratio of $\sim 2,165:1$ prior to pre-processing (for summary statistics, see Table 1). Of the 11 `int64` features, 3 are integer-encoded categorical IDs (`admission_type_id`, `discharge_disposition_id`, `admission_source_id`) with sentinel codes encoding missing values (Table 2), while the remaining 8 are cardinal numeric features (healthcare utilisation measures and length-of-stay) treated as continuous throughout, the majority exhibiting right-skewed distributions (Fig. 1). The 36 categorical features include demographics, administrative and diagnosis codes, clinical test results, and medication dosage indicators. Several exhibit high class imbalance: for example, 74.78% of encounters are from Caucasians (`race`) and 84.31% from patients over 50 (`age`), while “No [dose administered]” accounts for >90% of entries for 19 of the 23 dosage indicators (Fig. 2). Missing or “None” values are present for 13 features (all categorical or integer-encoded) – with five exhibiting >30% missingness (Fig. 3a) – while cardinality >15 is observed for `diag_1/2/3`, `medical_specialty`, and `payer_code` (Fig. 3b). The three-class target is imbalanced: only 11.2% of encounters are readmitted within 30 days (<30), compared to 35.0% after 30 days (>30) and 53.8% with no readmission (Fig. 4).

1.2 Challenges

(i) Missing data: ‘None’ values in `max_glu_serum` (94.7%) and `A1Cresult` (83.3%) are clinically informative, reflecting an active decision to not administer a test rather than missing data. Similarly, `diag_2` (0.4%) and `diag_3` (1.4%) missingness is structural rather than random: absent diagnoses reflect `number_diagnoses` < 2 or 3. For the remaining features with missing data, Little’s MCAR test ($\chi^2=1272.6$, $p<0.001$) rejects MCAR globally, ruling out simple imputation. Bonferroni-corrected logistic regressions of missingness indicators on observed features [2], alongside Welch’s *t*-tests across five clinical proxies, suggest that `weight` (96.9%) and `race` (2.2%) are likely MNAR,¹ while the remaining features appear consistent with MAR mechanisms (effect sizes $|d|<0.15$). This informs the imputation strategies in Table 3. **(ii) Patient-level leakage:** 101,766 encounters originate from 76,099 unique patients (up to 41 per patient), such that encounter-level splitting risks within-patient correlation inflating performance estimates (Fig. 5). **(iii) Class imbalance:** While the initial target is highly imbalanced, the requested binarisation of the target results in only mild imbalance (Fig. 4). **(iv) Near-zero Variance:** Twelve features (all medication indicators) exhibit near-zero variance (> 99.9% belonging to ‘No’; Fig. 5). **(v) Encoding complexity:** Three diagnosis fields contain 717–790 unique ICD-9 codes per field; codes carry hierarchical clinical meaning (e.g. 4xx = circulatory) that neither raw integer nor one-hot encoding preserve, while one-hot encoding would produce $\sim 2,100$ sparse columns. If not dropped, `medical_specialty` (73 categories) and `payer_code` would also require encoding (Fig. 3b).

Task 2: Data Assembling & Pre-processing

2.1 Assembly

Encounters with `discharge_disposition_id` indicating death – where readmission is impossible – or hospice transfer – where readmission represents goal-of-care reversal rather than routine

clinical readmission² – are excluded (Table 4), leaving 99,343 encounters. The target is binarised (<30, >30 $\rightarrow$ 1; No $\rightarrow$ 0) to give a 52.9:47.1 class ratio (Fig. 6). `patient_nbr` is retained as a grouping key for patient-level CV integrity (§3.3) then removed as a predictive feature to prevent identity leakage. A stratified 20,000-encounter subset is drawn, preserving class proportions and retaining 18,036 unique patients (Fig. 6). $N=20,000$ was selected to reduce nested CV compute cost while retaining sufficient sample diversity; the learning curve (Fig. 9) confirms CV AUC plateaus at $N\approx 9,900$, well before this threshold, validating that the reduced dataset retains the full predictive structure of the complete sample. Mild class imbalance is addressed via `class_weight='balanced'`, reweighting the loss function without discarding data or miscalibrating class priors. The `race` vector is preserved separately for a later fairness check (§4.3).

2.2 Pre-processing

Pre-pipeline cleaning recodes “missing” sentinel integer codes to NaN, ‘?’ strings to NaN – including `diag_1` (0.02%, $n=21$) – and `max_glu_serum` and `A1Cresult` ‘None’ values to `Not_Tested`. `diag_2` (0.4%) and `diag_3` (1.4%) NaN values are replaced with ‘None’ (Table 3). All remaining pre-processing is embedded within a `ColumnTransformer` inside each pipeline, fitted only on training partitions to prevent leakage. `weight` (96.9%), `medical_specialty` (49.1%), and `payer_code` (39.6%) are dropped due to high missingness, along with the twelve dosage indicators with near-zero variance (>99.9% No) as they contribute minimal discriminative signal even within CV folds (Fig. 5). This yields a final modeling dataset of 20,000 samples and 36 input features. For `race` (likely MNAR), mode imputation with a missingness indicator is used as a conservative approach, as the MNAR mechanism cannot be formally confirmed and the indicator preserves the signal; remaining MAR features also receive mode imputation within each fold (Table 3), again with missingness indicators as masking has been observed to improve predictive performance in health databases regardless of mechanism [3]. `diag_1` receives mode imputation only, as a missingness indicator would exhibit near-zero variance. Mode is used throughout as all imputed features are categorical. `age` is ordinal encoded (ten ordered brackets); `gender`, `change`, and `diabetesMed` are binary-encoded (0/1), implicitly mode-imputing `gender`’s 3 Unknown/Invalid values; and diagnosis features are encoded via ICD-9 clinical categories (§2.3). The remaining features receive ordinal encoding. While less problematic for tree-based models, one-hot encoding (OHE) would substantially increase dimensionality for LinearSVC without adding meaningful signal; target encoding is avoided as it introduces leakage risk within CV folds. While ordinal encoding introduces artificial metric distances for LinearSVC, this can be mitigated through regularisation (§3.4). `StandardScaler` is applied uniformly to maintain comparable feature scales for LinearSVC regularisation; tree-based models are unaffected but incur no penalty. Log-transforming right-skewed utilisation features was considered but not applied, as tree-based models are invariant to monotone transformations and applying different preprocessing per model would complicate comparison.

2.3 Diagnosis Encoding

Raw ICD-9 codes impose false ordinality on clinically unordered diagnostic identifiers; ordinal encoding inherits the same issue. OHE would introduce high dimensionality through $\sim 2,100$ sparse binary columns, while target encoding risks instability for rare codes alongside leakage within CV folds. We instead map each field to 13 ICD-9 clinical categories by disease system (Table 5) [4], preserving hierarchical clinical meaning at manageable car-

¹Weight-recorded patients had more lab procedures ($d=0.54$); race-missing patients had fewer prior inpatient visits ($d=0.33$).

²The readmission rate after hospice transfer is 11.2%, against 45.0% for home discharge.

dinality. The 13th category, `No_Diagnosis`, encodes structural absence in `diag_2/3` as a distinct value rather than conflating it with genuinely unclassified codes. All three fields are mapped independently, preserving the primary/secondary diagnostic distinction as separate features.

Task 3: Machine Learning Pipeline

3.1 Metrics

Primary (threshold-independent): ROC-AUC measures discrimination across all thresholds, enabling comparison with published readmission models where the equivalent c-statistic is the standard reported metric [5]. PR-AUC measures positive-class performance directly and is reported alongside ROC-AUC to provide a clinically grounded view of model utility: the clinical objective centres on correctly identifying readmissions rather than overall discrimination, and PR-AUC is sensitive to false positives in a way that ROC-AUC is not. **Secondary (threshold-dependent, $\tau=0.5$):** Recall is prioritised due to the higher cost of missed readmissions; Precision, F1, and Balanced Accuracy are additionally reported. Clinically motivated threshold optimisation is analysed in §3.6. **Calibration:** Brier Score assesses probability faithfulness (≤ 0.25 = no-skill baseline at 47% prevalence), important for actionable risk communication. Plain accuracy is excluded as models use `class_weight= 'balanced'`, making raw accuracy a misleading performance measure.

3.2 Models

Three models are compared: a linear baseline and two ensemble methods representing distinct variance-reduction paradigms; both ensemble methods have demonstrated strong performance on this dataset relative to linear baselines [6]. (i) **Linear Support Vector Machine (SVM):** `LinearSVC` wrapped in `CalibratedClassifierCV` (sigmoid, 3-fold) for probability outputs. Provides an interpretable linear decision boundary and a performance lower bound for non-linear models. (ii) **Random Forest (RF)** (bagging): averages independent decision trees to reduce variance while capturing non-linear interactions common in tabular clinical data. (iii) **Gradient Boosting (GB)** (boosting): sequentially fits trees to residual errors, capturing complex relationships via a complementary ensemble paradigm to RF.

3.3 Cross-Validation

Nested cross-validation is used to optimise hyperparameters while providing unbiased estimates of test performance. Each pipeline consists of: (i) a `ColumnTransformer` — mode imputation with missingness indicators, encoding, `StandardScaler` — followed by (ii) a classifier, cloned and refitted independently within each fold to prevent data leakage. **Outer loop (test performance):** `StratifiedGroupKFold` ($k=5$, `shuffle=True`, `seed=42`) with `groups=patient_nbr` ensures all encounters from the same patient appear exclusively in either the training or test set (80:20 outer split) while preserving the class distribution across folds, preventing both patient-level leakage and fold imbalance. **Inner loop (hyperparameter selection):** Within each outer training fold, `GridSearchCV` with `StratifiedGroupKFold` ($k=3$, `seed=42`) selects hyperparameters by maximising ROC-AUC, using the search spaces described in §3.4. Repeat encounters are retained as this reflects real deployment settings where hospitals predict readmission for patients with prior admissions; however, within-fold patient correlation from repeat encounters means estimates may represent a slight upper bound on generalisable performance.

3.4 Hyperparameter Analysis

Search spaces (inner CV): SVM $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$; RF $n_{est} \in \{100, 200\}$, $depth \in \{5, 10\}$, $min_leaf \in \{1, 5\}$, $max_features \in \{\sqrt{\cdot}, \log_2\}$ (16 candidates); GB $n_{iter} \in \{100, 200\}$, $lr \in \{0.05, 0.1, 0.2\}$, $depth \in \{3, 5\}$ (12 candidates). Fig. 8 plots CV and train ROC-AUC across all combinations from the 20,000-sample `GridSearchCV` refit. For **SVM** (Fig. 8a), CV AUC is flat across all C (0.6567–0.6570) with a consistently small train–CV gap (≈ 0.006), confirming a high-bias regime: the linear boundary

cannot capture non-linear structure regardless of regularisation strength. Best $C=0.01$. For **RF** (Figs. 8b–e), $depth=10$ achieves higher CV AUC (0.675) than $depth=5$ (0.668) but at the cost of a markedly wider train–CV gap (0.122–0.158 vs 0.022–0.024). $min_samples_leaf=5$ partially mitigates this at $depth=10$ (gap: 0.158→0.122). $n_estimators$ and $max_features$ have negligible effects on CV AUC (ranges: < 0.001). Best: $depth=10$, $min_leaf=5$. For **GB** (Figs. 8f–h), $depth=3$ outperforms $depth=5$ across all learning rates (best CV AUC 0.676 vs 0.673), consistent with boosting’s weak-learner principle favouring shallow trees. At $depth=5$, CV AUC declines monotonically with learning rate (0.673→0.671→0.669) while the train–CV gap widens (0.060→0.068→0.080), indicating overfitting from overly complex trees with aggressive updates. At $depth=3$, $lr=0.1$ is optimal (0.6756); n_iter has negligible effect (0.6756 vs 0.6755). Best: $depth=3$, $lr=0.1$.

3.5-3.6 Performance Table and Discussion of Results

Table 1: Nested CV performance (mean $\pm$ SD, 5 outer folds). Val = inner CV mean; Test = outer fold mean. All metrics at $\tau=0.5$ except AUC, PR-AUC, and Brier. Best performers are shown in bold.

Model	Split	AUC	PR-AUC	Rec.	Prec.	F1	Bal.Acc	Brier
SVM	Val	.657 $\pm$.002	.632 $\pm$.005	.456 $\pm$.013	.622$\pm$.003	.526 $\pm$.009	.604 $\pm$.003	.230 $\pm$.000
	Test	.658 $\pm$.008	.633 $\pm$.016	.457 $\pm$.008	.623$\pm$.021	.527 $\pm$.012	.605 $\pm$.008	.230 $\pm$.001
RF	Val	.675$\pm$.003	.644$\pm$.005	.562 $\pm$.007	.611 $\pm$.005	.585$\pm$.006	.621$\pm$.003	.227$\pm$.001
	Test	.677$\pm$.011	.647$\pm$.017	.564 $\pm$.012	.615 $\pm$.018	.588$\pm$.015	.625$\pm$.009	.227 $\pm$.002
GB	Val	.672 $\pm$.003	.643 $\pm$.005	.565$\pm$.010	.605 $\pm$.007	.584 $\pm$.008	.618 $\pm$.004	.227$\pm$.001
	Test	.674 $\pm$.011	.647$\pm$.017	.569$\pm$.023	.605 $\pm$.022	.586 $\pm$.023	.619 $\pm$.014	.226$\pm$.002

RF achieves the highest test AUC (0.677 $\pm$ 0.011), marginally above GB (0.674 $\pm$ 0.011). Wilcoxon signed-rank tests are used for pairwise model comparison in place of parametric tests, as normality assumptions are unjustifiable at $n=5$ folds.[2] All pairwise comparisons reach significance at $\alpha=0.05$: RF vs. SVM, GB vs. SVM, and RF vs. GB each yield $W=15$, $p=0.031$, $r=1.00$, the minimum achievable p -value at $n=5$, indicating consistent directional ordering across all folds rather than formal statistical significance. Despite gradient boosting typically outperforming bagging on tabular data, RF marginally exceeds GB in the high-bias regime identified here, where GB’s sequential residual focusing offers no advantage over RF’s parallel variance reduction. RF is therefore selected as the best model for Task 5. The learning curve (Fig. 9) is evaluable from $N \approx 6,500$. CV AUC rises from 0.630 to 0.674 before plateauing, indicating a high-bias regime in which the performance ceiling is feature-determined rather than data-limited, consistent with Kansagara *et al.* (2011), who attribute similar AUC ceilings to the absence of post-discharge and social determinants data from administrative records.[5] Applying a cost-sensitive threshold of $\tau^*=0.320$ ($C_{FN}/C_{FP}=3$, illustrative, Fig. 10) raises Recall from 0.580 to 0.971 at the cost of reduced Precision (0.647→0.515), demonstrating that clinically motivated threshold selection can substantially increase sensitivity for patient flagging. Calibration (Fig. 11) improves SVM probability outputs, reducing the Brier score from 0.234 to 0.228 ($\Delta=0.007$), with the largest corrections occurring at the extremes of the predicted probability distribution.

Task 4: Model Interpretation

4.1 Feature Importance

Prior healthcare utilisation dominates importance rankings across all three models, with `number_inpatient` the top-ranked feature under every method. Fig. 12 presents native importance: absolute standardised coefficients for SVM, Gini impurity reduction for RF, and permutation importance (AUC-based) for GB. Fig. 13 presents SHAP `TreeExplainer` values for RF and GB, and `LinearExplainer` for SVM.[7] SHAP is preferred over Gini as Gini inflates importance for high-cardinality features; this is directly demonstrated in RF where `num_lab_procedures` ranks 4th by Gini (0.072) but 14th by SHAP (0.004), while `admission_source_id` rises from Gini rank 13 to SHAP rank 4. The SHAP dependence plot for `number_inpatient` (Fig. 14) reveals a threshold relationship: 66% of samples have negative

SHAP values (zero prior inpatient visits reduces predicted readmission probability), with values ranging from -0.076 to $+0.241$.

4.2 Cross-Model Comparison

By native importance, four features appear in the top 10 of all three models: `number_inpatient`, `number_diagnoses`, `number_outpatient`, and `number_emergency`; the SHAP consensus is broader at eight features, additionally including `age`, `admission_source_id`, `num_procedures`, and `diag_1`, with the discrepancy reflecting cardinality artefacts in native measures that SHAP corrects. **SVM** uniquely elevates missingness indicators (`admission_type_id_missing`, `admission_source_id_missing`, `race_missing`) into its top 10, suggesting these encode systematic administrative recording patterns with strong linear separability but negligible tree-model importance. **GB** SHAP concentrates importance more sharply (`number_inpatient`: 0.338 vs **RF** 0.067), consistent with sequential residual focusing. The absence of individual medication indicators from the SHAP consensus suggests treatment choices provide limited discriminative signal beyond disease burden and prior utilisation.

4.3 Clinical Interpretation

The consensus features are clinically plausible: `number_inpatient` and `number_emergency` capture repeated healthcare contact, consistent with the LACE index [8]; `number_diagnoses` reflects comorbidity burden; `admission_source_id` proxies acuity; and `diag_1` is expected given that readmission risk varies by condition. As these reflect prior utilisation and disease burden rather than modifiable targets, the model is better suited to risk stratification than identifying actionable interventions. Two features are harder to explain clinically. First, individual medication indicators are absent from the SHAP consensus despite this being a diabetes dataset, suggesting treatment choices provide limited signal once disease burden is accounted for. This may reflect treatment standardisation: once comorbidity burden and prior utilisation are established, medication choices largely follow clinical protocols rather than acting as independent risk signals; alternatively, near-zero variance removal of 12 dosage indicators (§2.2) may have suppressed medication signal before it reached the model. Second, SVM's elevation of missingness indicators (`admission_type_id_missing`, `race_missing`) into its top features is more plausibly explained by systematic administrative recording differences than genuine clinical risk. Two further limitations apply: utilisation features partially encode prior readmission history for repeat encounters; and `diabetesMed` reflects severity judgements made during the encounter rather than acting as an independent predictor. The fairness audit (Fig. 15, RF, equalised-odds framework) shows near-parity for the two largest groups (AfricanAmerican AUC 0.683 vs Caucasian 0.671), with violations in smaller groups: Asian TPR 0.385 ($\Delta=-0.174$ vs overall) and Other violating both TPR and FPR ($\Delta=-0.164$, $\Delta=-0.101$), though estimates are unreliable at these sample sizes (Asian $n=120$, Other $n=295$).

Task 5: Alternative Pipeline

5.1 Pipeline Design

The Task 3 pipeline is modified to incorporate the best model from §3.6, **RF**, with an embedded feature selection step. **Strategy A** constitutes the primary modified pipeline: `SelectFromModel` (RF, `threshold $\tau$`) is fitted exclusively within each training partition to prevent selection-bias leakage. The threshold $\tau \in \{\text{'median'}, \text{'mean'}, \text{'1.25*mean'}\}$ is tuned jointly with RF hyperparameters in the inner CV (`StratifiedGroupKFold`, $k=3$, `groups=patient_nbr`), so threshold selection cannot observe test data. All other components are identical to Task 3. At $\tau=\text{'median'}$ (selected in all 5 folds) exactly 18 of 36 features are retained. As an additional extension, **Strategy B** evaluates `RFECV`(`LinearSVC`, `step=1`, `scoring='roc_auc'`), which

cross-validates the optimal feature count via iterative coefficient-magnitude elimination — a fundamentally different selection criterion (linear sparsity vs. non-linear importance) applied to the SVM baseline. Features stable across both strategies and both algorithmic paradigms can then be identified as a model-agnostic robust core with the strongest claim to genuine predictive signal, independent of selection criterion. One limitation applies: `RFECV` does not support group-aware CV, so `StratifiedKFold` is used in the inner loop; the negligible val-to-test AUC gap ($\Delta=-0.001$) suggests patient leakage in inner folds has minimal practical effect.

5.2–5.3 Results, Discussion, and Impact of Feature Selection

Table 2: Modified pipeline performance (mean $\pm$ SD, 5 folds). Val=inner CV mean; Test=outer fold mean. Baselines from §3.5 included for reference. Best per metric in bold.

Pipeline	Sp.	AUC	PR-AUC	Rec.	Prec.	F1	BA	Brier	Feats	Jacc.
RF (36)	Val	675$\pm$.003	644$\pm$.005	562 $\pm$.007	611 $\pm$.005	585$\pm$.006	621$\pm$.003	.227 $\pm$.001	36	—
	Test	677$\pm$.011	647$\pm$.017	564 $\pm$.012	615 $\pm$.018	588$\pm$.015	625$\pm$.009	.227 $\pm$.002	36	—
SVM (36)	Val	.657 $\pm$.002	.632 $\pm$.005	.456 $\pm$.013	622$\pm$.003	.526 $\pm$.009	.604 $\pm$.003	.230 $\pm$.000	36	—
	Test	.658 $\pm$.008	.633 $\pm$.016	.457 $\pm$.008	623$\pm$.021	.527 $\pm$.012	.605 $\pm$.008	.230 $\pm$.001	36	—
A: SFM+RF	Val	.672 $\pm$.003	.639 $\pm$.005	.555 $\pm$.003	.607 $\pm$.006	.580 $\pm$.004	.617 $\pm$.003	227$\pm$.001	18 $\pm$ 0	.937 $\pm$.052
	Test	.674 $\pm$.009	.642 $\pm$.016	.557 $\pm$.012	.611 $\pm$.017	.583 $\pm$.014	.621 $\pm$.007	227$\pm$.002	18 $\pm$ 0	.937 $\pm$.052
B: RFECV+SVM	Val	.659 $\pm$.003	.634 $\pm$.005	.458 $\pm$.013	.624 $\pm$.002	.528 $\pm$.010	.606 $\pm$.003	.229 $\pm$.001	19 $\pm$ 3	.752 $\pm$.065
	Test	.658 $\pm$.009	.633 $\pm$.016	.461 $\pm$.008	.625 $\pm$.020	.531 $\pm$.012	.608 $\pm$.008	.230 $\pm$.001	19 $\pm$ 3	.752 $\pm$.065
Δ AUC A vs RF: -0.003 Δ AUC B vs SVM: $+0.000$ Cross-strategy Jacc.: 0.459 ± 0.044										

Strategy A achieves test AUC 0.674 $\pm$ 0.009 ($\Delta=-0.003$ vs full RF), confirming that discarding half the features causes negligible performance loss in the high-bias regime identified in §3.6. **Strategy A** achieves perfect hyperparameter stability (`threshold=median` in all 5 folds) and near-perfect feature stability (Jaccard=0.937 $\pm$.052, 17 of 18 features selected in every fold), indicating a robust, deployment-confident feature set. **Strategy B** matches the full SVM (Δ AUC= $+0.000$) with moderate feature stability (Jaccard=0.752 $\pm$.065, 19.4 $\pm$ 3.4 features per fold), lower than **Strategy A**, reflecting that `RFECV` optimises feature count independently per fold rather than applying a fixed threshold. The low cross-strategy overlap (Jaccard = 0.459 $\pm$.044) confirms that importance-based and sparsity-based criteria are complementary; the 7 features stable across both strategies (`number_inpatient`, `number_diagnoses`, `number_outpatient`, `num_procedures`, `age`, `diag_1`, `admission_source_id`) converge with the SHAP consensus from §4.2, providing the strongest evidence of genuine predictive signal. **Strategy B** additionally stabilises all four missingness indicators, consistent with their linear separability noted in §4.2, while `num_lab_procedures` appearing in **Strategy A**'s stable set despite ranking 14th by SHAP directly demonstrates the Gini cardinality bias discussed in §4.1. Halving the feature set improves clinical interpretability and reduces inference cost without sacrificing discrimination, though it does not resolve the fairness disparities identified in §4.3, as structural inequities in the training data are independent of feature count.

Several limitations apply across the full pipeline. Within-fold patient correlation from repeat encounters (§3.3) means performance estimates may represent a slight upper bound on generalisable performance. The performance ceiling is feature-determined rather than data-limited (§3.6), as administrative records lack post-discharge and social determinants data. **Strategy B**'s non-group-aware inner CV (§5.1) and unreliable fairness estimates for underrepresented racial groups (§4.3) further limit the strength of conclusions in those specific analyses.

References

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